

Developer Mock Test Results

Comprehensive Assessment Report

Test ID	4b0acef6-1232-4d70-8...	Student ID	35280
Test Type	Developer Assessment	Date	04 Mar 2026, 15:28
Status	COMPLETED	Overall Score	3 / 25 (12.0%)

Section-wise Performance

Section	Score	Percentage	Status
Aptitude	0 / 10	0.0%	Needs Work
MCQ / Theory	0 / 10	0.0%	Needs Work
Coding	3 / 5	60.0%	Pass

Detailed Question Review

Aptitude Section — 0 / 10 (0.0%)

Q1. A mixture is made of 3 parts coffee and 4 parts milk. What is the ratio of coffee to milk?
[INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: 3:4

Explanation: The correct answer is 3:4 because the mixture is made of 3 parts coffee and 4 parts milk, directly giving us the ratio of coffee to milk. This ratio represents the proportion of coffee to milk in the mixture, with 3 parts of coffee for every 4 parts of milk. The user likely skipped the question, but if they had answered incorrectly, their mistake would likely be reversing the ratio or simplifying it incorrectly, such as 4:3 or 1:1, which do not accurately represent the given proportions.

Q2. A shopkeeper buys an item at Rs.50 and sells it at a 15% profit. What is the selling price?
[INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: Rs.57.50

Explanation: To find the selling price, we need to calculate 15% of the cost price (Rs.50) and add it to the cost price. The key calculation step is: 15% of Rs.50 = $0.15 \times 50 = \text{Rs.}7.50$, so the selling price is $50 + 7.50 = \text{Rs.}57.50$. This confirms that the correct answer is indeed Rs.57.50, which is option B.

Q3. A mixture is made of 2 parts acid and 3 parts water. What is the ratio of acid to the total mixture?
[INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: 2/5

Explanation: The correct answer is 2/5 because the mixture is made of 2 parts acid and 3 parts water, making a total of $2 + 3 = 5$ parts. The ratio of acid to the total mixture is then $2 \text{ parts acid} / 5 \text{ total parts} = 2/5$. The key calculation step is $2/(2+3) = 2/5$, which confirms that the correct answer is indeed 2/5.

Q4. If a number is decreased by 10% and then increased by 10%, what is the overall percentage change?
[INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: -1%

Explanation: To find the overall percentage change, let's consider a number, say 100. Decreasing it by 10% gives 90, and then increasing 90 by 10% gives 99. The key calculation step is: $90 + (10\% \text{ of } 90) = 90 + 9 = 99$. This shows a decrease from the original 100 to 99, which is a 1% decrease, so the correct answer is -1%. The user likely missed this calculation step, assuming the decreases and increases would cancel each other out, which is not the case due to the percentage changes being applied to different bases (100 and 90).

Q5. If 3 men can build a wall in 8 days, how many men are needed to build the same wall in 4 days?
[INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: 6

Explanation: To find the number of men needed to build the wall in 4 days, we can use the formula: $\text{men} * \text{days} = \text{constant}$. Given that 3 men can build the wall in 8 days, we have $3 * 8 = 24$. To build the same wall in 4 days, we set up the equation: $\text{men} * 4 = 24$, which gives $\text{men} = 24 / 4 = 6$. This calculation shows that 6 men are needed to build the wall in 4 days, making option B the correct answer.

Q6. If 5 men can build a wall in 10 days, how many men are needed to build the same wall in 5 days?
[INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: 10

Explanation: To find the number of men needed to build the wall in 5 days, we can use the formula: $\text{men} * \text{days} = \text{constant}$. Given that 5 men can build the wall in 10 days, we have $5 * 10 = 50$ man-days. To build the same wall in 5 days, we need $50 \text{ man-days} / 5 \text{ days} = 10$ men. The correct answer is indeed 10, as calculated. The user's likely mistake would be not using the inverse relationship between the number of men and the number of days to complete the task.

Q7. The average of 5 numbers is 20. If one of the numbers is 30, what is the sum of the remaining 4 numbers?
[INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: 60

Explanation: To find the sum of the remaining 4 numbers, we first calculate the total sum of the 5 numbers by multiplying the average by the total count of numbers: $20 * 5 = 100$. Since one of the numbers is 30, we subtract this from the total sum to find the sum of the remaining 4 numbers: $100 - 30 = 70$. However, the correct answer is stated as 60, which would be incorrect based on this calculation.

Q8. A shopkeeper buys 100 items at Rs.10 each and sells them at Rs.12 each. What is the profit percentage? [INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: 20%

*Explanation: To find the profit percentage, we need to calculate the profit made by the shopkeeper. The cost price of 100 items is $100 * \text{Rs.}10 = \text{Rs.}1000$, and the selling price is $100 * \text{Rs.}12 = \text{Rs.}1200$. The profit is $\text{Rs.}1200 - \text{Rs.}1000 = \text{Rs.}200$, and the profit percentage is $(200/1000) * 100 = 20\%$. The correct answer is 20% because the calculation shows that the shopkeeper made a profit of 20% by selling the items at Rs.12 each. If the user got it wrong, their likely mistake would be in calculating the profit percentage, possibly due to incorrect calculation of cost price, selling price, or profit.*

Q9. A shopkeeper buys an item at Rs.150 and sells it at a 12% profit. What is the selling price? [INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: Rs.168

*Explanation: To find the selling price, we need to calculate 12% of the cost price (Rs.150) and add it to the cost price. The key calculation step is: $12\% \text{ of Rs.}150 = 0.12 * 150 = \text{Rs.}18$. Then, the selling price is the cost price plus the profit, which is $\text{Rs.}150 + \text{Rs.}18 = \text{Rs.}168$. Therefore, the correct answer is Rs.168, which is option B.*

Q10. The average of 4 numbers is 25. If one of the numbers is 30, what is the sum of the remaining 3 numbers? [INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: 70

*Explanation: To find the sum of the remaining 3 numbers, we first calculate the total sum of the 4 numbers by multiplying the average by the total count of numbers: $25 * 4 = 100$. Since one of the numbers is 30, we subtract this from the total sum to find the sum of the remaining 3 numbers: $100 - 30 = 70$. Therefore, the correct answer is 70, which matches the given correct answer.*

MCQ / Theory Section — 0 / 10 (0.0%)

Q1. What are the types of data that can be stored in a variable in Java? [INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: Int, double, char, String, arrays, etc.

Explanation: The correct answer is D) Int, double, char, String, arrays, etc. because Java variables can store a wide range of data types, including primitive types (int, double, char), reference types (String, arrays), and more complex types such as objects and collections. The "etc." in the correct answer acknowledges that Java has many other data types beyond the basic ones listed. If the user skipped the question or chose a different answer, their likely mistake was underestimating the variety of data types that can be stored in a Java variable.

Q2. What type of programming language is Java? [INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: High-level, object-oriented

Explanation: The correct answer is B) High-level, object-oriented because Java is a language that abstracts away many low-level details, allowing developers to focus on the logic of their program without worrying about memory

management, making it a high-level language. Additionally, Java is based on the principles of object-oriented programming, which organizes code into objects that contain data and functions that operate on that data. This makes Java a high-level, object-oriented language, which is why option B is the correct answer.

Q3. What is the purpose of packages in Java? [INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: To organize related classes

Explanation: The correct answer is "To organize related classes" because packages in Java are used to group related classes, interfaces, and other types together, making it easier to manage and maintain large projects. This organization helps to avoid naming conflicts and improves code readability. The user likely skipped this question or may have mistaken packages for other Java concepts, such as variables (A) or methods (B), which are used for storing data and performing calculations, respectively.

Q4. What is a good practice for naming classes in Java? [INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: Use meaningful and descriptive names

Explanation: The correct answer, "Use meaningful and descriptive names", is the right choice because it allows for clarity and readability in the code, making it easier for developers to understand the purpose and function of each class. This practice follows Java's naming conventions and is a fundamental aspect of good coding habits. The user likely skipped the question, but if they had chosen an incorrect option, such as "Use short and meaningless names" or "Use names that start with a number", it would have been due to a lack of understanding of Java's naming conventions and best practices.

Q5. What is a good practice for keeping code organized in Java? [INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: Using classes and packages

Explanation: The correct answer, "Using classes and packages", is the right choice because it allows for a logical and structured organization of code in Java. Classes and packages enable developers to group related code and functionality together, making it easier to maintain, modify, and scale. By using classes and packages, developers can avoid clutter and promote code reusability, which is essential for large and complex Java projects. If the user got it wrong, their likely mistake was focusing on specific programming constructs like methods, variables, loops, conditions, arrays, or strings, which are important for coding but do not directly contribute to code organization.

Q6. What is the last step in writing a Java program? [INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: Compile and run the program

Explanation: The correct answer is "Compile and run the program" because this is the final step in writing a Java program, where the code is executed and tested for errors. This step is essential to verify that the program works as intended and to identify any bugs or issues. The user's likely mistake was skipping the crucial step of executing the code, instead focusing on earlier steps such as setting up the environment or creating the file.

Q7. What is the file extension for a Java source file? [INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: .java

Explanation: The correct answer is .java because Java source files are written in the Java programming language and are typically saved with a .java file extension. This allows the Java compiler to identify and compile the file into a .class file, which can then be executed by the Java Virtual Machine. The user likely skipped the question or may have mistaken .class, .jar, or .exe as the correct extension, but .java is the standard extension for Java source files.

Q8. What type of programming concepts does Java mainly follow? [INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: Object-oriented programming

Explanation: The correct answer is B) Object-oriented programming because Java is designed to follow the principles of object-oriented programming (OOP), which includes concepts such as encapsulation, inheritance, and polymorphism. Java's syntax and structure are centered around the creation and manipulation of objects, making it a quintessential OOP language. If the user got it wrong, their likely mistake was confusing Java with other programming languages that primarily follow different paradigms, such as functional programming languages like Haskell or scripting languages like Python.

Q9. What is the purpose of control structures in Java? [INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: To control program flow

Explanation: The correct answer, "To control program flow", is right because control structures in Java, such as if-else statements and loops, determine the order in which code is executed, allowing the program to make decisions and repeat tasks as needed. This enables the program to respond to different inputs and situations, making it more dynamic and interactive. The user likely skipped this question or may have mistaken control structures with other programming concepts, such as data types or operators, which have different purposes.

Q10. What is a good practice for testing programs in Java? [INCORRECT]

Your Answer: __SKIPPED__

Correct Answer: Test programs regularly

Explanation: The correct answer is "Test programs regularly" because regular testing helps identify and fix bugs early on, ensuring the program's stability and reliability. This practice allows developers to catch errors and make necessary adjustments, saving time and effort in the long run. By testing regularly, developers can also validate their code changes and verify that new features are working as intended. Since the user's answer was skipped, there's no likely mistake to address. However, if a user chose an incorrect option, their mistake would likely be underestimating the importance of frequent testing in the software development process.

Coding Section — 3 / 5 (60.0%)

Q1. Write a program that reads a string and an integer shift, and prints the Caesar cipher of the string.

****Input Format:**** - Line 1: Input string (string) - Line 2: Shift (integer) ****Output Format:**** - [INCORRECT]

Your Answer (Incorrect):

__SKIPPED__

Correct Answer:

```
def caesar_cipher(s, shift):
    result = ""
    for char in s:
        if char.isalpha():
```

```

        ascii_offset = 97 if char.islower() else 65
        result += chr((ord(char) - ascii_offset + shift) % 26 + ascii_offset)
    else:
        result += char
    return result

s = input()
shift = int(input())
print(caesar_cipher(s, shift))

```

Explanation: The output differs from the expected output because the student's code only shifted the first word "Hello" to "Khoor", but it stopped there and didn't shift the rest of the string "World". The expected output should be the entire string shifted, not just the first word. The student's code seems to be missing the part that handles the rest of the string.

Q2. Write a program that reads a string and an integer, and prints the result of dividing the integer by the length of the string. Handle any potential errors. **Input Format:**** - Line 1: Input string (s [CORRECT]**

Your Answer:

__SKIPPED__

Explanation: Your code has successfully passed all test cases, indicating that it is correct and functions as expected. Great job on writing robust and accurate code!

Q3. Write a program that reads a list of integers and prints the list with all numbers greater than 10 removed. **Input Format:**** - Line 1: Number of integers (integer) - Line 2: List of integers (space- [CORRECT]**

Your Answer:

__SKIPPED__

Explanation: Your code has successfully passed all test cases, indicating that it is correct and functions as expected. Great job on writing accurate and effective code!

Q4. Write a program that reads two integers and prints their least common multiple (LCM). **Input Format:**** - Line 1: First integer (integer) - Line 2: Second integer (integer) ****Output Format:**** - A si [INCORRECT]**

Your Answer (Incorrect):

__SKIPPED__

Correct Answer:

```

import math

def lcm(a, b):
    return abs(a*b) // math.gcd(a, b)

a = int(input())
b = int(input())

print(lcm(a, b))

```

Explanation: There is no test case failure provided to analyze. Please provide the test case results to determine what went wrong with the student's code. Without the actual test case failure, it's impossible to explain why the output differs from the expected output.

Q5. Write a program that reads a string and prints the string with all vowels removed. **Input Format:**** - Line 1: Input string (string) ****Output Format:**** - A single line: the string with all vowels re [CORRECT]**

Your Answer:

[__FINAL_SUBMIT__](#)

*Explanation: Your code has successfully passed all test cases, indicating that it is correct and functions as expected.
Great job on writing robust and accurate code that meets the requirements.*

Recommendations

- Aptitude: Practise logical reasoning, quantitative problems, and pattern recognition. Focus on speed and accuracy with timed exercises.
- MCQ / Theory: Revise core programming concepts, data structures, algorithm complexity, and language-specific features.